KASARANENI UPESH CHOWDARY

AI Engineer | Generative AI | Agentic AI | LLM Applications | Computer Vision

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PROFESSIONAL SUMMARY

Computer Science Engineering student (CGPA 8.56, 7th semester) focused on AI Engineering, Generative AI, agentic systems and applied machine learning. Builds end-to-end AI products with Python, FastAPI, LLM APIs, multimodal AI, computer vision and workflow automation. Experienced in multi-agent orchestration, agent memory, structured outputs, tool calling, grounded generation, evaluation, hallucination mitigation, API integration and production-oriented data pipelines. Interested in building reliable AI systems that turn real-world data into actionable workflows with human-in-the-loop decision support.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL
Generative AI / Agents: LLMs, AI Agents, Multi-Agent Systems, Agentic Workflows, RAG, LangChain, Prompt Engineering, Structured Outputs, Tool Calling, Agent Memory, MCP, Multimodal AI, Vision-Language Models, NLP
Reliability / Evaluation: Grounding, Hallucination Mitigation, LLM Evaluation, Precision, Recall, F1, Cohen's Kappa, Uncertainty Handling, Deterministic Rules Engines, Guardrails, Fail-Open Design, Human-in-the-Loop, Decision Logging, Audit Trails, Observability, Provider Abstraction
Vector / ML: Vector Databases, Embeddings, FAISS, ChromaDB, Sentence-Transformers, PyTorch, TensorFlow, scikit-learn, Transformers, Computer Vision, YOLOv8, OpenCV, OCR, Barcode Recognition, Perceptual Hashing, Regression
LLM APIs / Data: Gemini API, Claude API, OpenAI API, Groq API, Ollama, Whisper, Tavily, SearXNG, FastAPI, REST APIs, Pydantic, PostgreSQL (Row-Level Security), Supabase, MySQL, MongoDB, SQLite
Frontend / DevOps: React, Vite, Tailwind CSS, Framer Motion, Three.js, Recharts, Radix UI, Lucide React, Responsive Design, CSS 3D Transforms, Scroll-triggered Animations, Docker, pytest, Git, GitHub, Render, Vercel
Automation / Product: n8n, Antigravity, Claude Code, Cursor, Kimi CLI, ChatGPT, Go-to-Market (GTM), E-commerce / Amazon Seller Workflows, Reverse Logistics

PROJECTS

Sydon Returns - AI Returns Manager for Amazon Sellers (CUBE Buildathon 2026)
Sep 2026 - Oct 2026

Python, Gemini (Vision LLM), FastAPI, Pydantic, PostgreSQL, Supabase, OpenCV, ZXing, scikit-learn, MCP, React, TypeScript, Three.js, Docker, Render, pytest

- Built a vision-LLM agent that inspects returned-parcel photos against the seller's catalogue, order record and parts list to decide identity, completeness, Amazon condition grade and disposition (restock / refurbish / liquidate / dispose), recovering margin lost when resellable returns are liquidated.
- Kept decisions out of the LLM: Gemini returns one batched, Pydantic-validated structured judgment per return (tool calling, ≤2 round trips), and a deterministic 24-rule disposition engine with human sign-off gates for dispose, high-value and escalated cases decides the outcome.
- Treated UNCERTAIN as a first-class verdict: identity is confirmed only by matched product features or a barcode read (ZXing + OpenCV), images are perceptual-hashed, and model or quota failures fail open to human review instead of a guessed grade; 85% confidence gate for auto-approval.
- Engineered for production: Postgres row-level-security multi-tenancy, JWT / API-key auth, job queue with circuit breaker and quota guard, SHA-256 hash-chained evidence records for the downstream Recovery agent and an MCP server; 558 passing pytest tests, deployed on Render.

CommerceSense AI - AI-Powered Amazon Seller Copilot
Sep 2026

Python, Gemini, Ollama, LangChain, FAISS, ChromaDB, Tavily, SearXNG, FastAPI, React, TypeScript, SQLite, scikit-learn, Sentence-Transformers, Docker, pytest

- Architected a multi-agent commerce pipeline where an Orchestrator coordinates Research and Insight agents across Listing, Pricing, Review and Inventory Intelligence modules.
- Implemented deterministic-first analytics for inventory risk, PPC waste, rank / Buy Box signals, listing health, pricing, review themes and demand forecasting, keeping business metrics outside the LLM.
- Built a RAG pipeline with LangChain that indexes Sentence-Transformers (Transformers) embeddings of reviews and listings in FAISS / ChromaDB vector databases, grounding LLM insights in retrieved evidence.
- Added structured outputs, web-search API integration, caching, decision logging, CSV import validation, feature flags and observability; containerized with Docker and validated by 107 passing pytest tests; audit evaluation over 14 products detected 9/9 planted signals with 0 false alarms.
- Built pluggable Gemini/Ollama and Tavily/SearXNG provider layers with embedding-based semantic review clustering (Sentence-Transformers + scikit-learn) and deterministic fallback when optional LLM services are unavailable.

JARVIS - Multimodal AI Personal Assistant
Mar 2026 - Present

Python, Claude API, Groq API, OpenAI API, Ollama, OpenCV, NumPy, OCR, Whisper, n8n, REST APIs

- Engineered a multimodal AI assistant combining LLM reasoning, conversational agent memory, screen understanding, screenshot analysis and OCR for context-aware interaction.
- Integrated 4 LLM providers with Whisper speech-to-text, OpenCV and n8n through API integrations to support agentic workflows for voice, vision and desktop task automation.
- Designed modular AI, vision and OCR components for maintainable, reliable tool-driven execution and extensible computer-control workflows.

StockWise AI - AI-Powered Stock Prediction Platform
Mar 2026 - Present

Python, NumPy, Machine Learning, Spring Boot, ReactJS, MySQL, Yahoo Finance API

- Built an end-to-end market analytics platform applying predictive machine-learning models to Yahoo Finance data for stock trend forecasting and regression-oriented analysis.
- Developed data-processing workflows, Spring Boot REST APIs with JWT authentication, React dashboards and MySQL storage for portfolio and prediction insights; validated APIs with Postman and deployed on Render and Vercel.

EDUCATION

Koneru Lakshmaiah Education Foundation (KLEF) - B.Tech, Computer Science and Engineering
Jul 2023 - May 2027

Vijayawada, Andhra Pradesh | CGPA: 8.56 | 7th Semester

Srichaithanya Jr College - Intermediate (MPC)
Aug 2021 - May 2023

Vijayawada, Andhra Pradesh | CGPA: 7.50

Nalanda High School - 10th Grade
Aug 2020 - Apr 2021

Edupugallu, Andhra Pradesh | CGPA: 7.52

CERTIFICATIONS & ACHIEVEMENTS

- Microsoft Certified: Azure Fundamentals (AZ-900).
- Oracle Redwood Application 2025 Certified Developer Associate - Oracle.
- Linguaskill General - Cambridge English.
- Solved 150+ Data Structures and Algorithms problems on LeetCode.